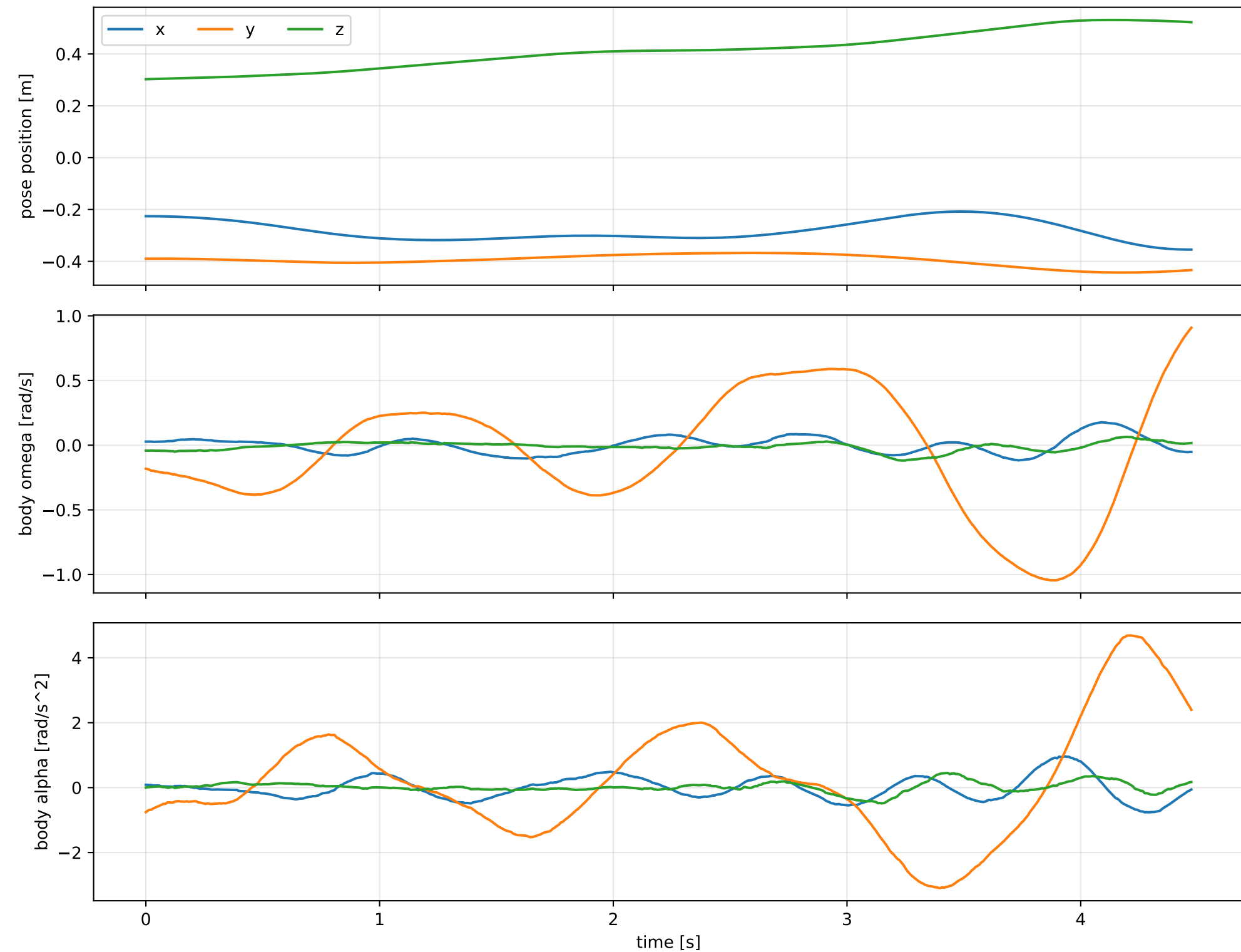
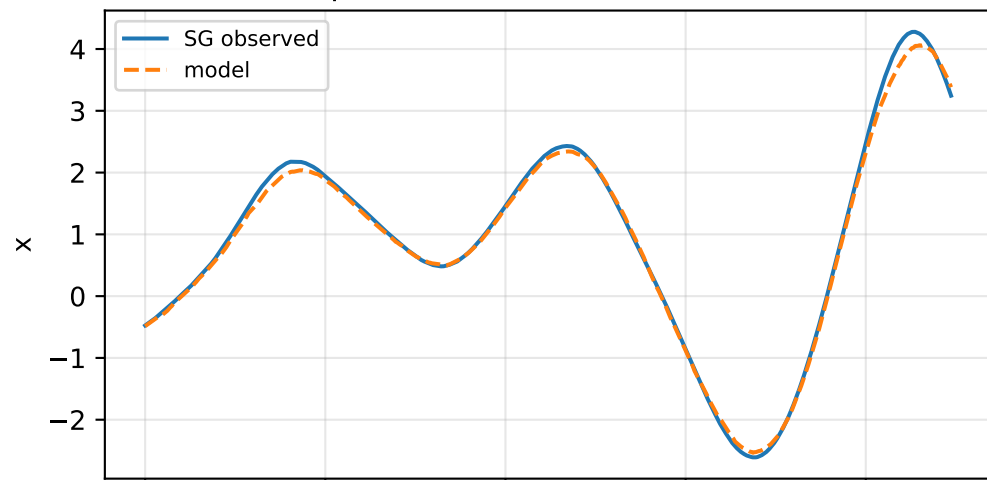


kkt_scale_negative: SG trajectory and derived motion

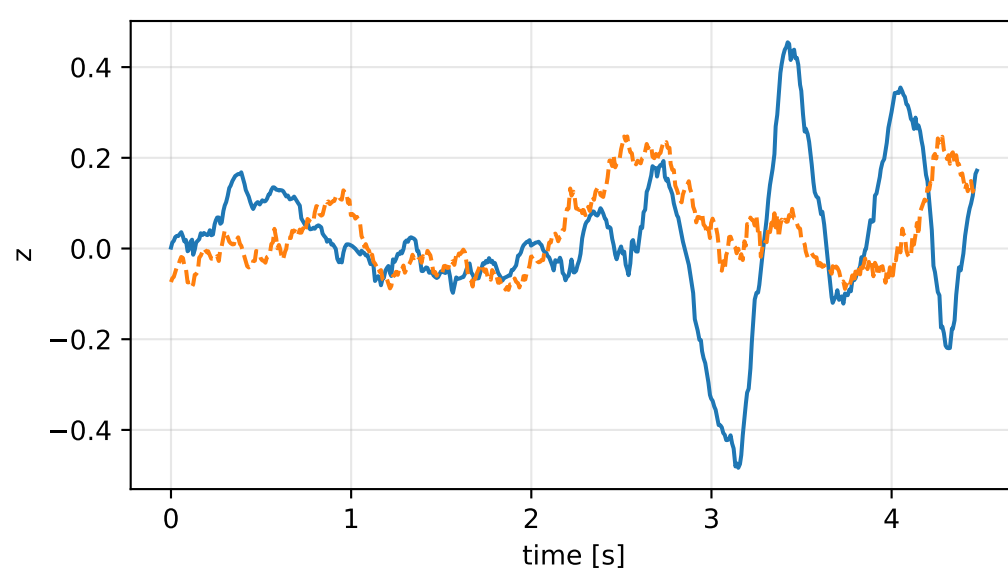
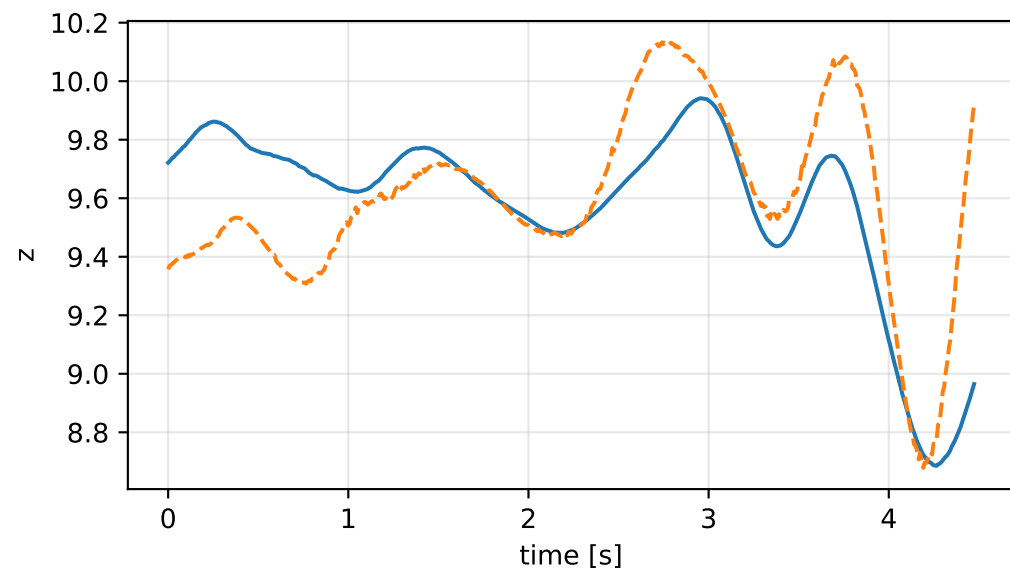
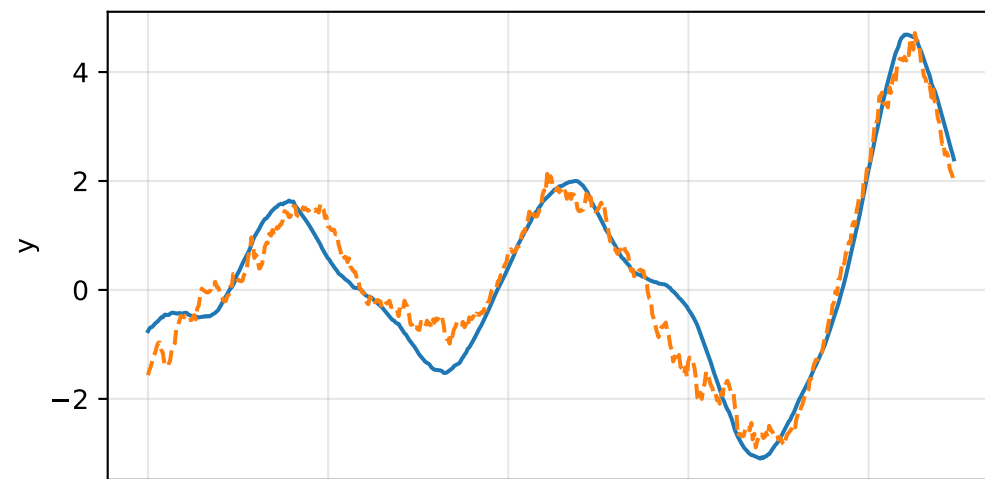
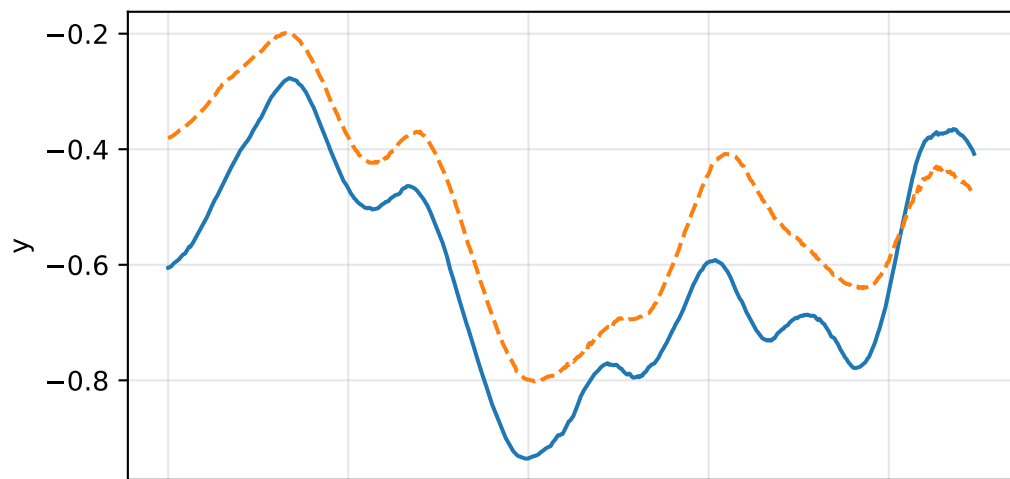
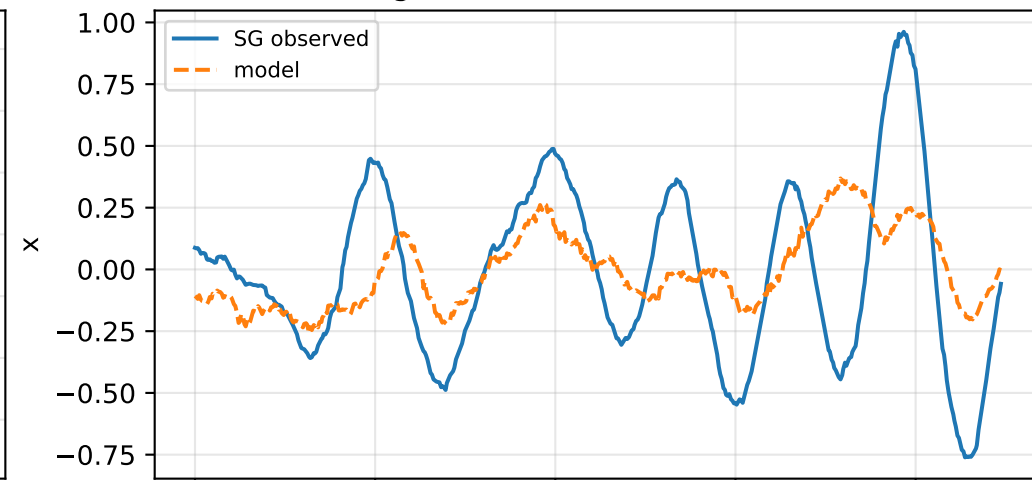


kkt_scale_negative: acceleration residual objective

specific acceleration [m/s^2]

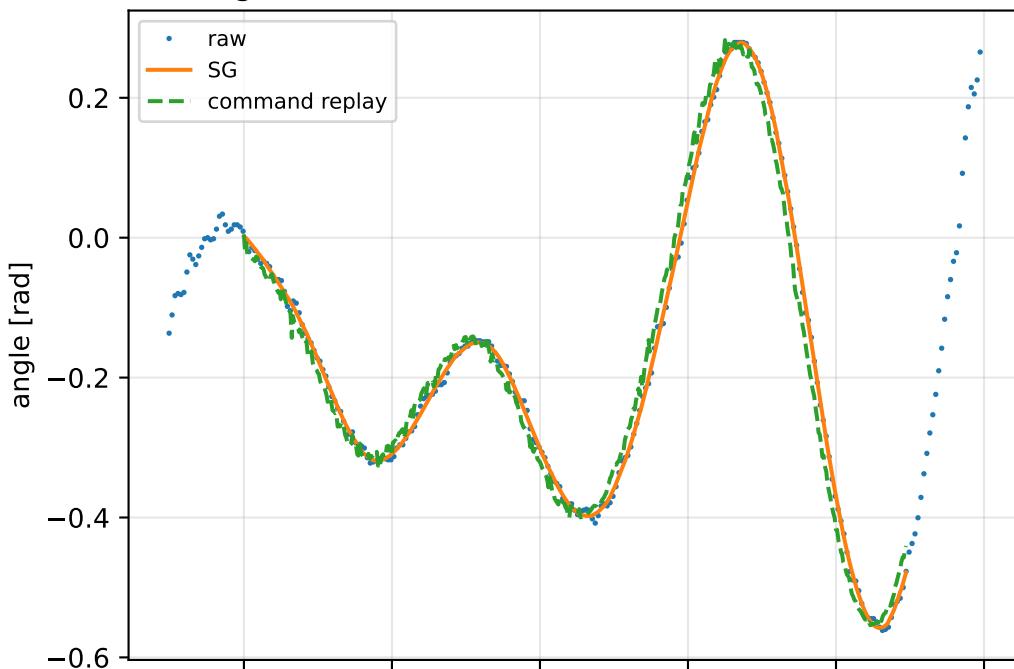


angular acceleration [rad/s^2]

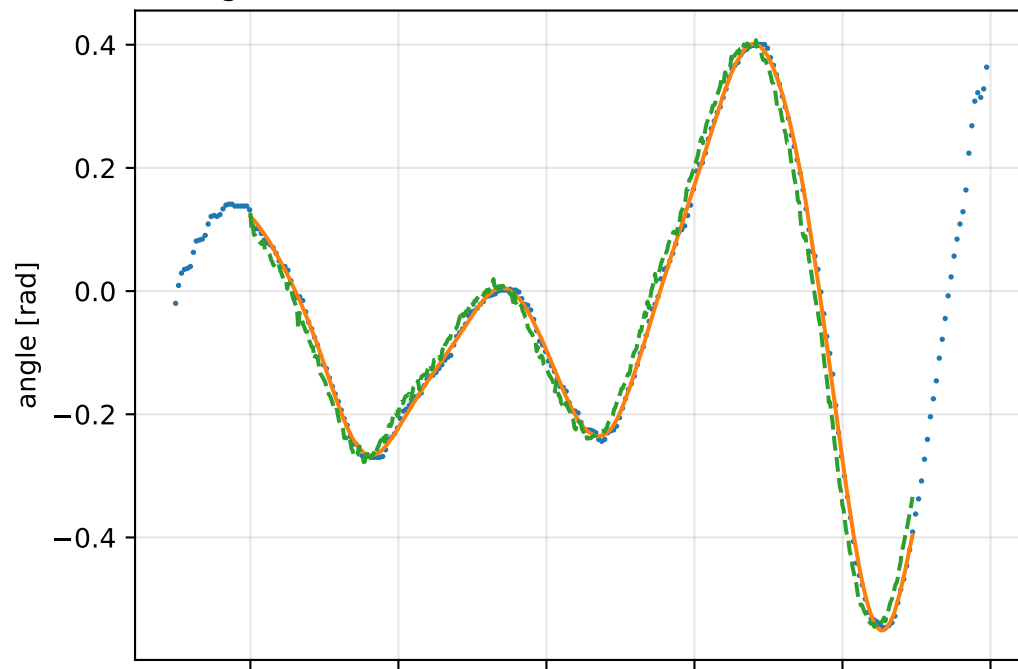


kkt_scale_negative: gimbal measurement audit (objective=measured_sg)

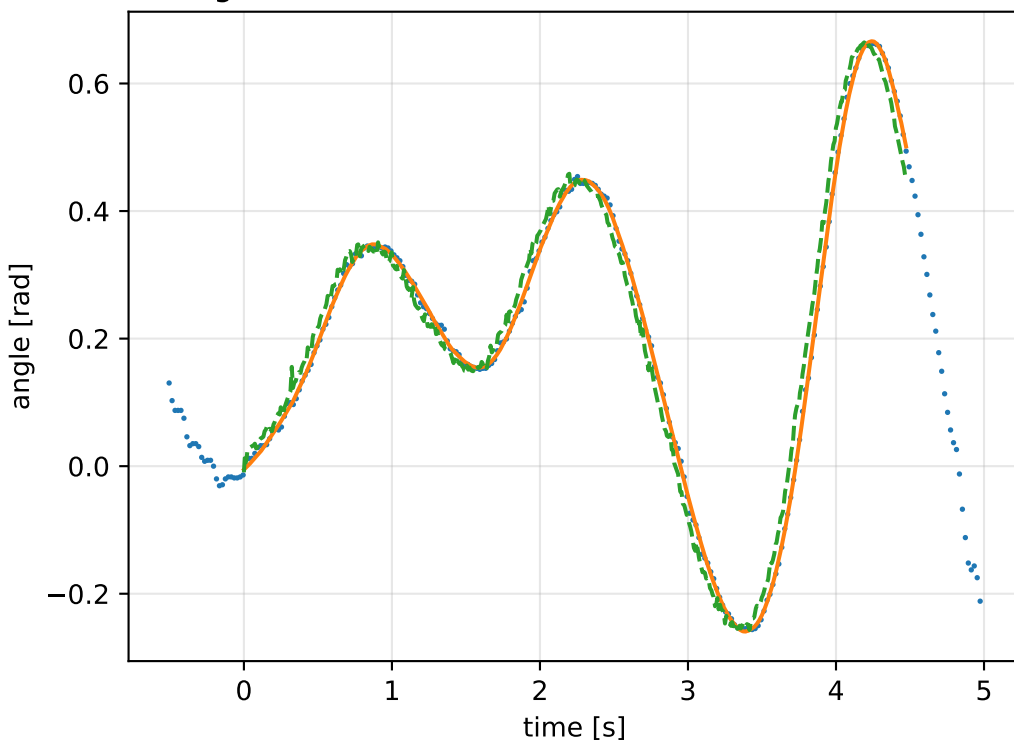
gimbal 1: RMSE=0.027, max=0.07529 rad



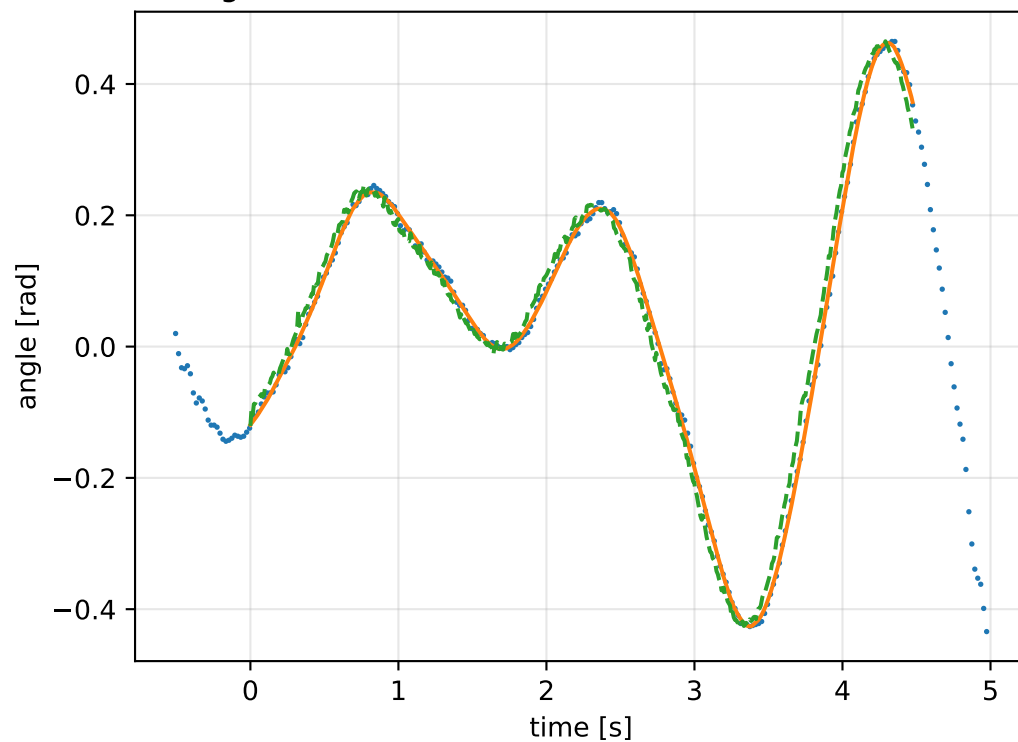
gimbal 2: RMSE=0.03107, max=0.08239 rad



gimbal 3: RMSE=0.03106, max=0.08682 rad

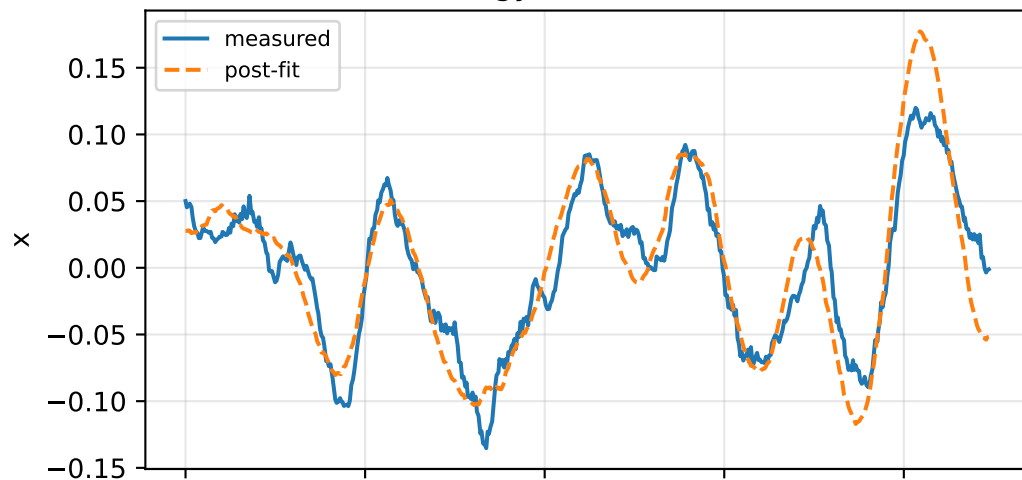


gimbal 4: RMSE=0.02598, max=0.07261 rad

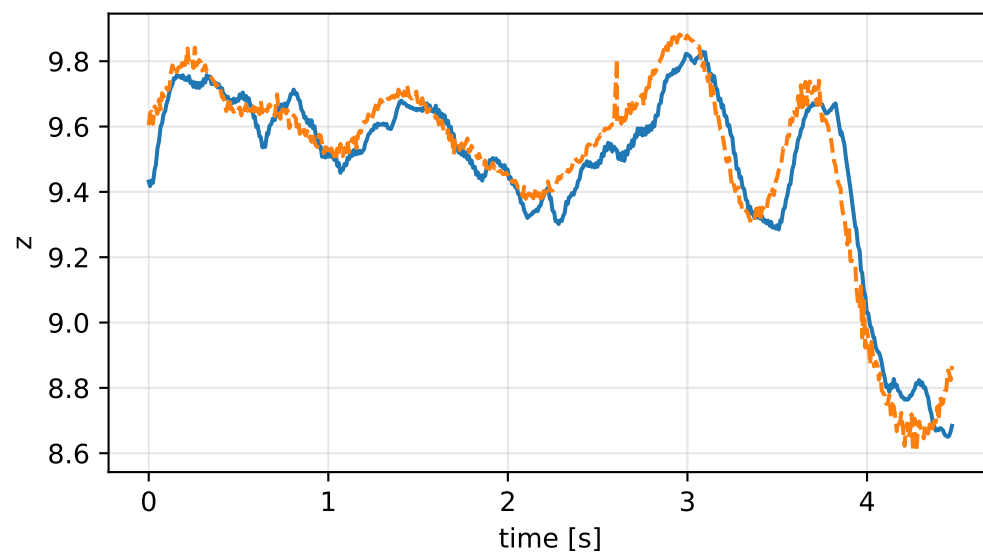
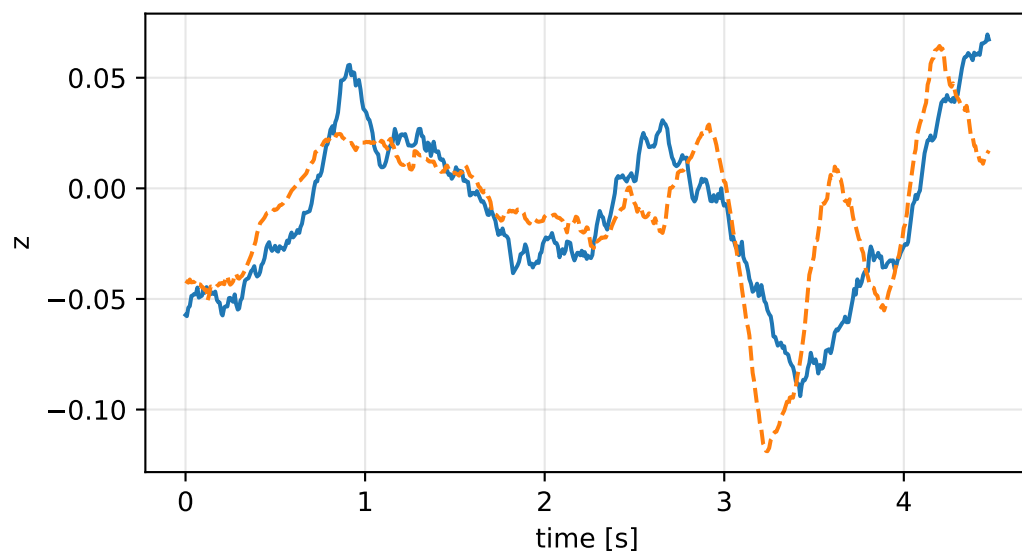
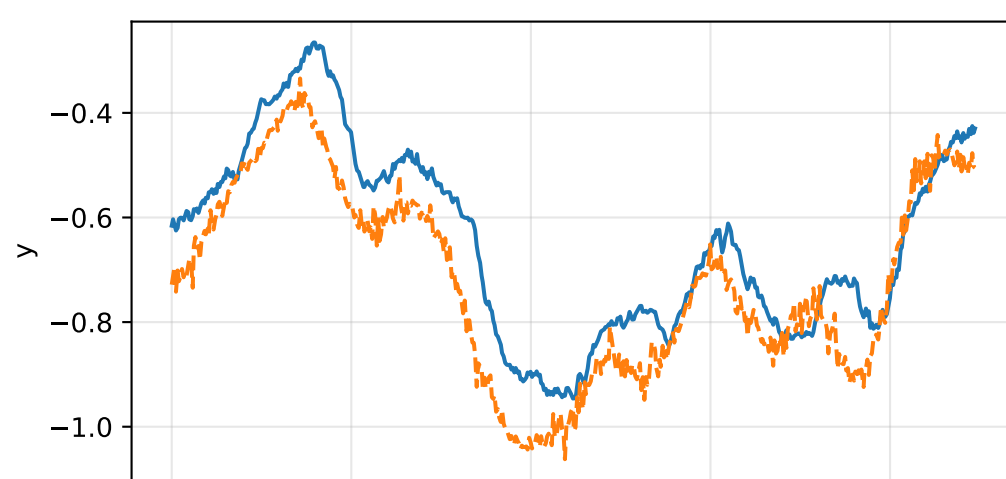
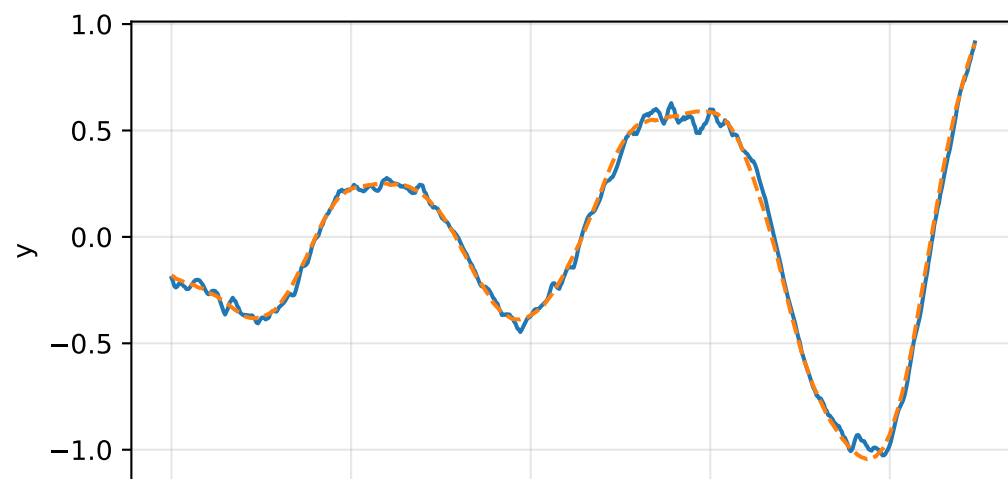
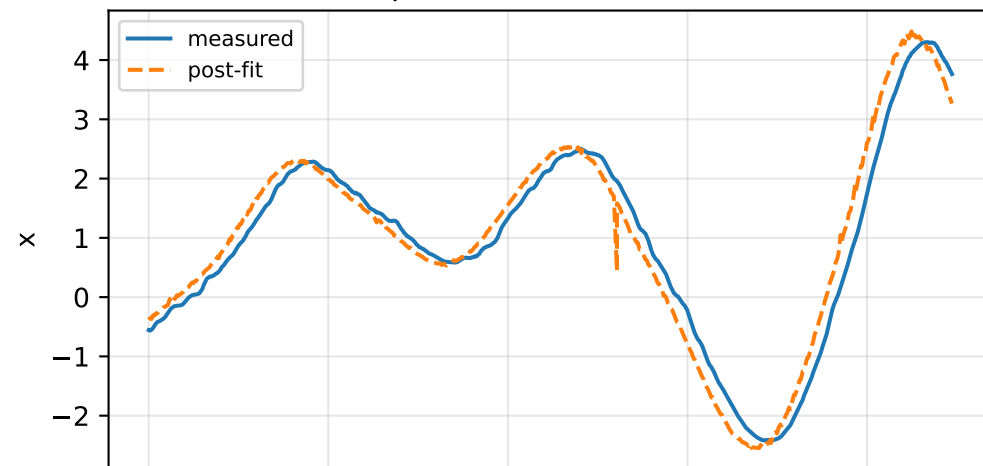


kkt_scale_negative: IMU comparison (diagnostic only)

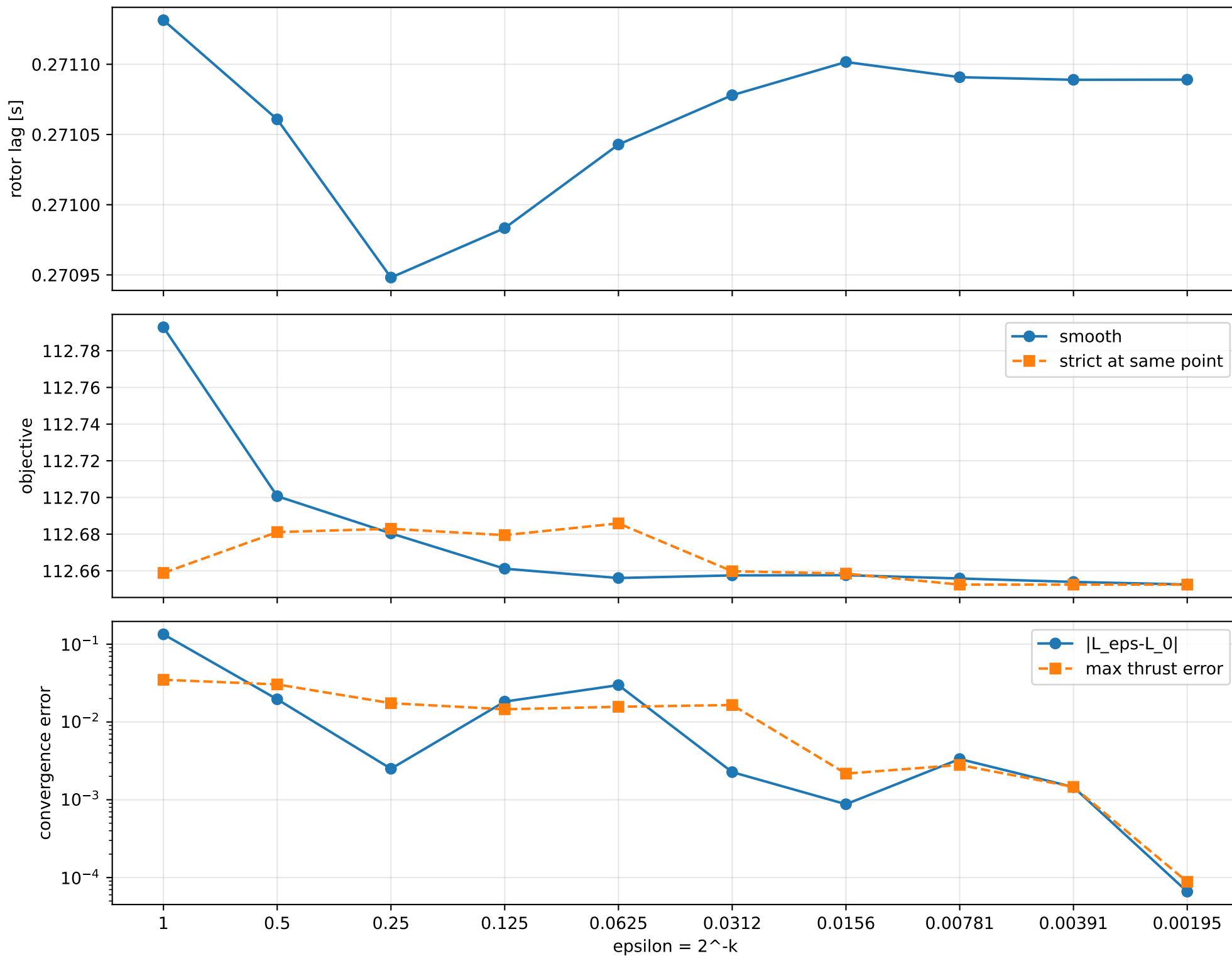
gyro [rad/s]



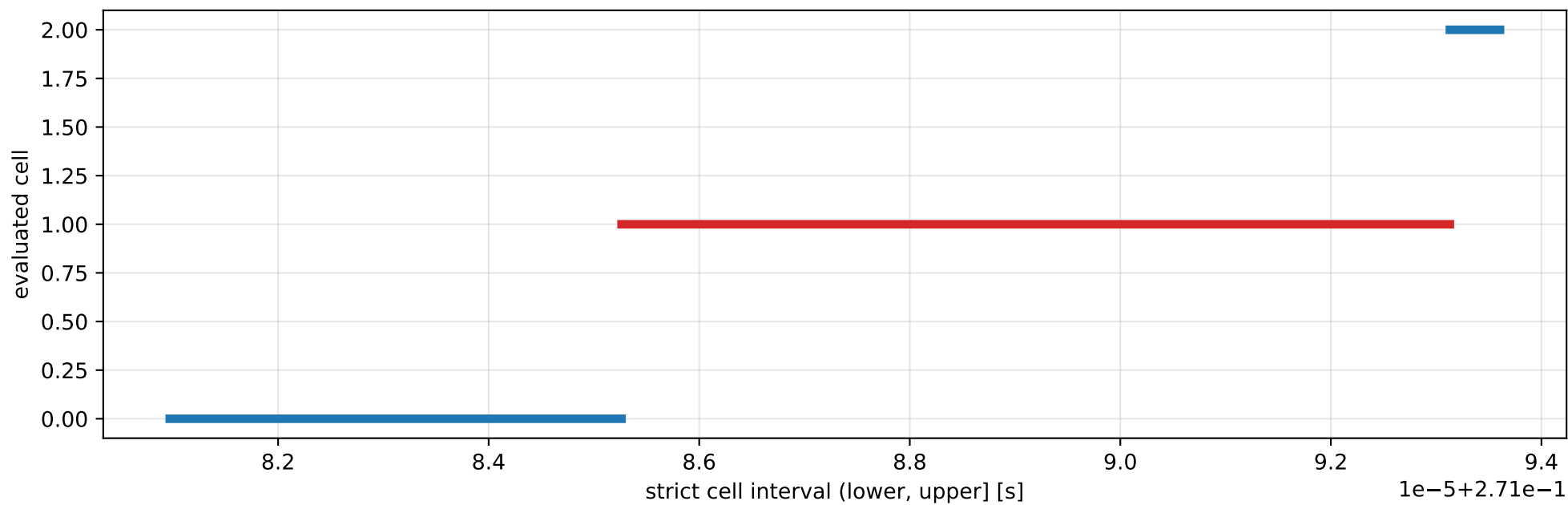
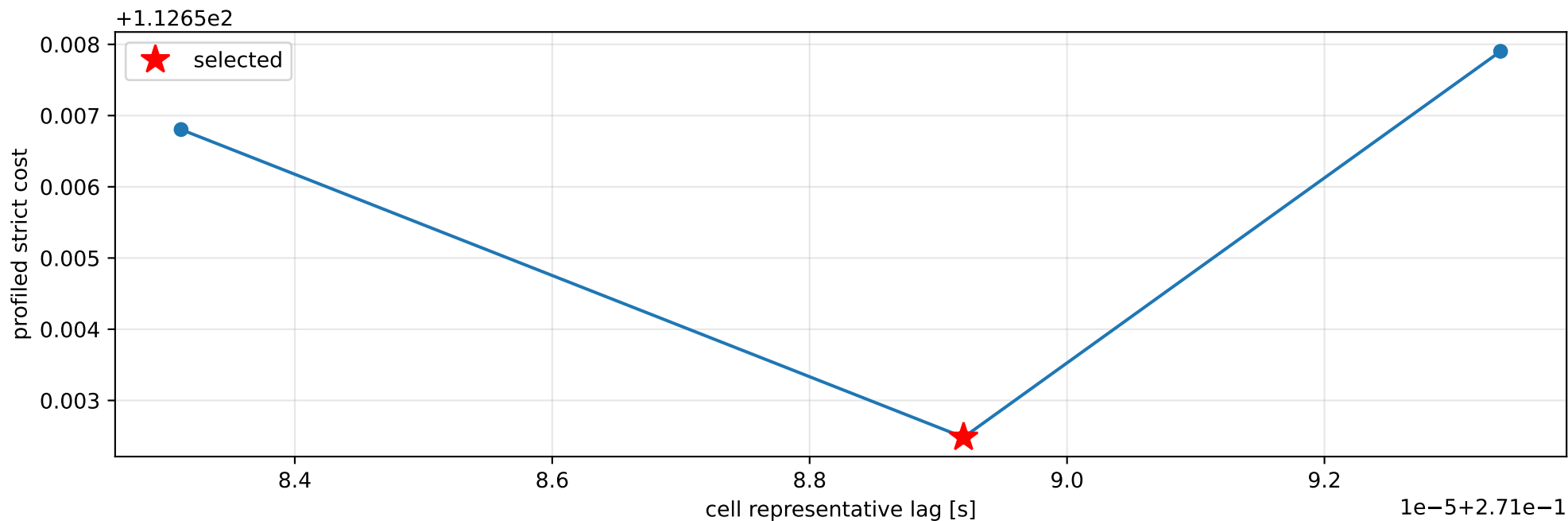
specific force [m/s^2]



kkt_scale_negative: smooth-to-strict continuation



kkt_scale_negative: exact strict-ZOH lag cells



selected (0.271085262, 0.27109313] s; final smooth 0.271089061 s; data boundary=False

kkt_scale_negative: parameter estimate

Case: kkt_scale_negative

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 7.52043166

inertia [kg m²]:

```
[[ 0.55443706  0.01372501 -0.00323582]
 [ 0.01372501  0.26168334 -0.08172244]
 [-0.00323582 -0.08172244  0.79125002]]
```

CoG body [m]: [-0.026435 0.01133929 0.00941535]

force effectiveness: [2.87131359 2.32387844 2.56341471 3.07927639]

scale-free J/m [m²]:

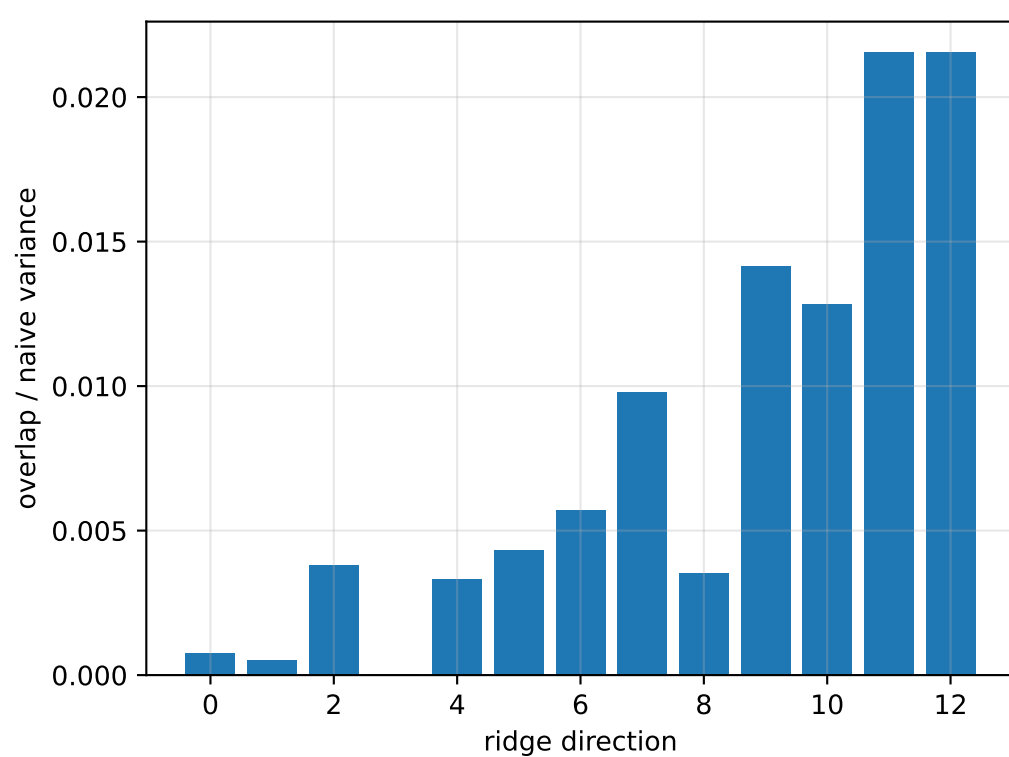
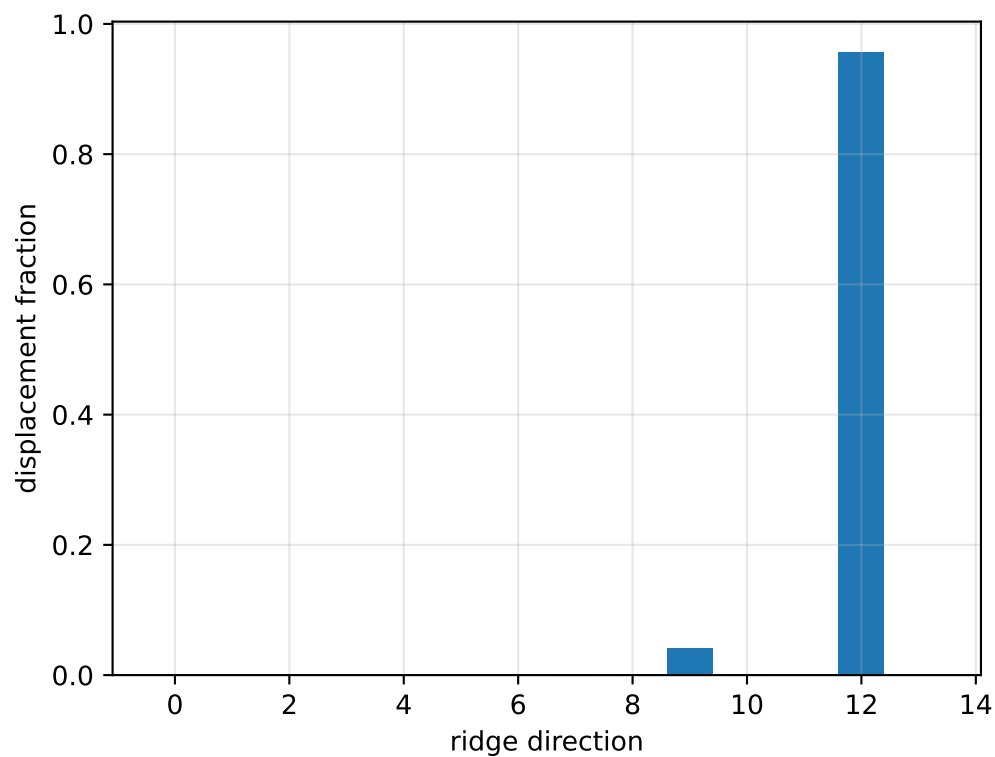
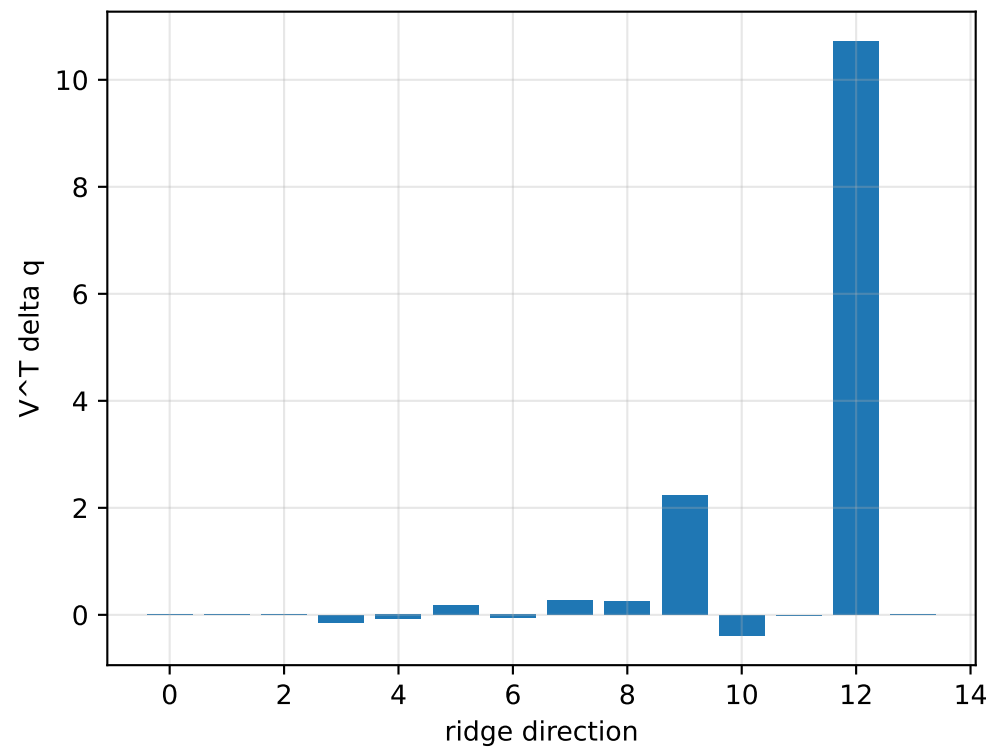
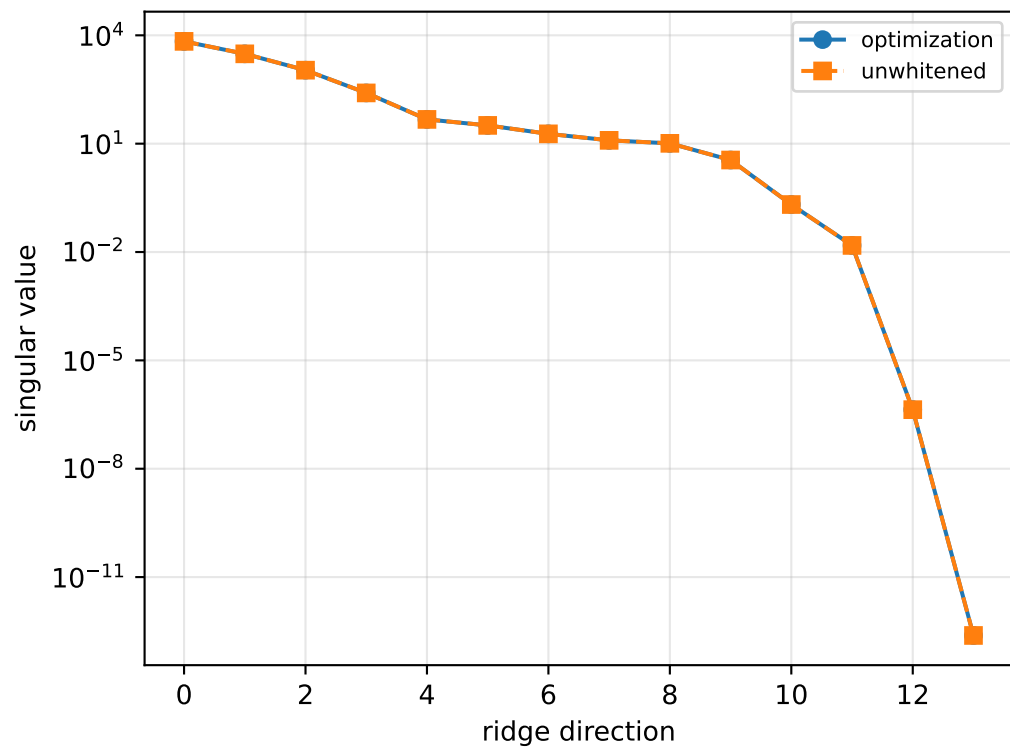
```
[[ 0.0737241  0.00182503 -0.00043027]
 [ 0.00182503  0.03479632 -0.01086672]
 [-0.00043027 -0.01086672  0.10521338]]
```

scale-free f/m: [0.3818017 0.30900865 0.34086005 0.40945474]

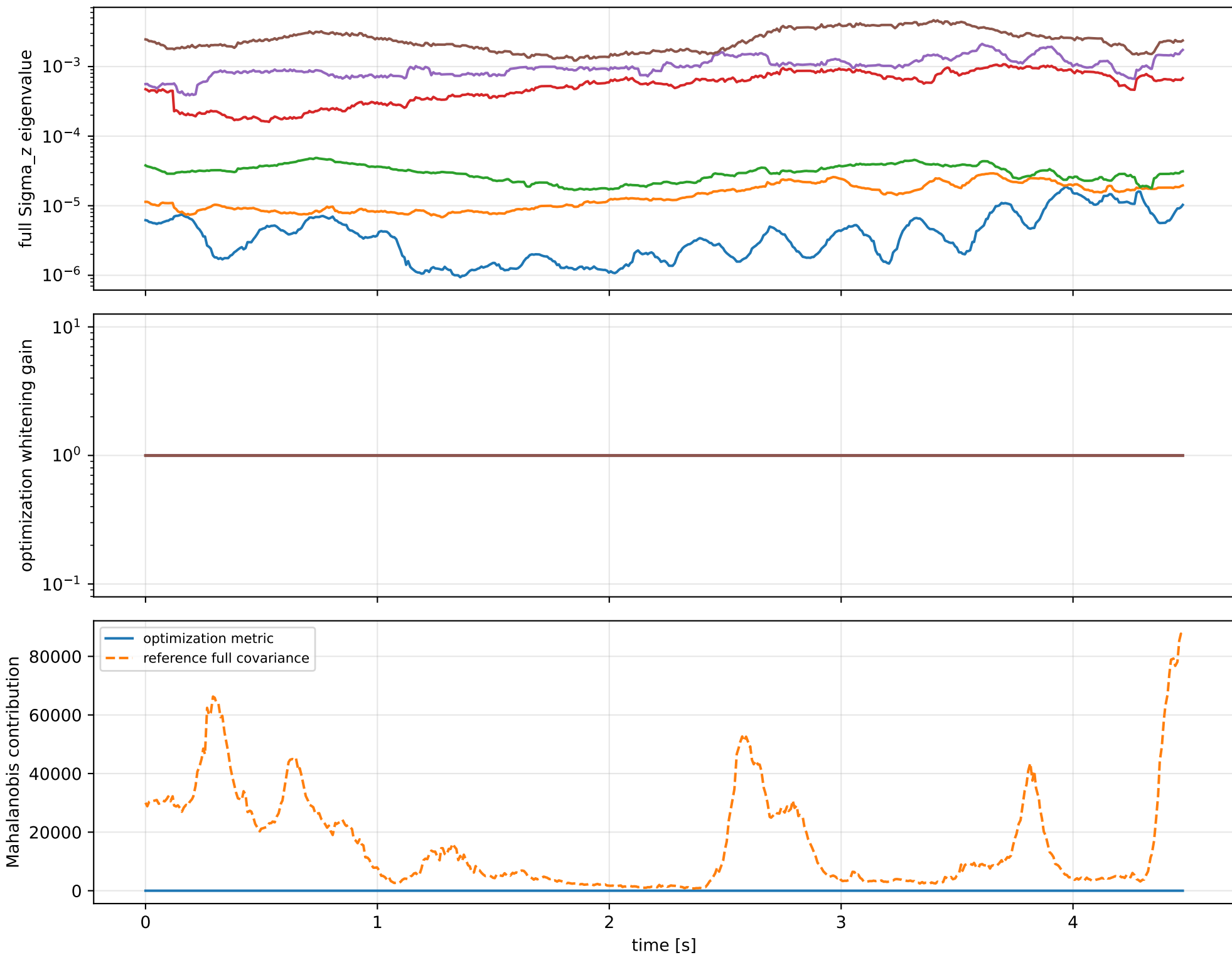
rotor lag cell: (0.271085262, 0.27109313] s

representative: 0.271089196 s

kkt_scale_negative: ridge spectrum (rank 13, nullity 1, $||jv||=1.56e-13$)



kkt_scale_negative: optimization vs reference covariance



kkt_scale_negative: wrench / closure (force max $2.24\text{e-}14$, torque max $3.89\text{e-}16$)

